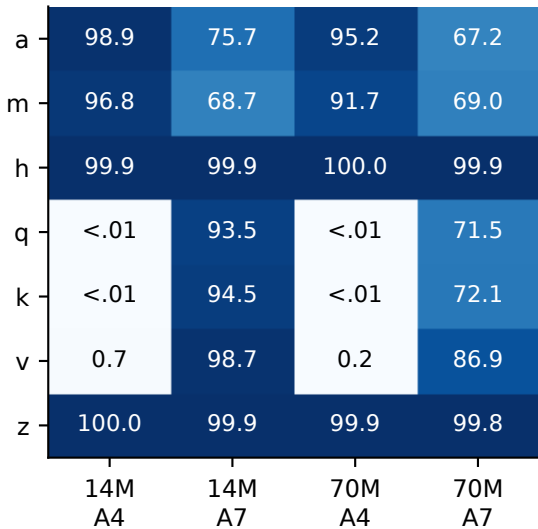
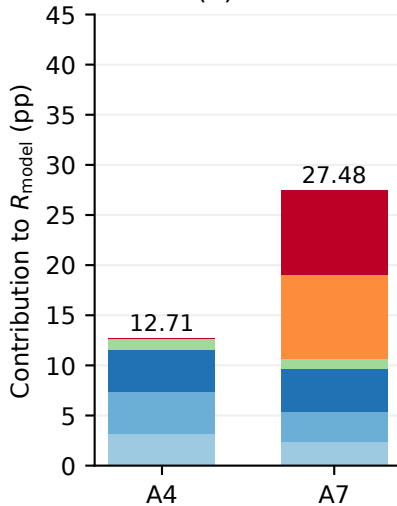


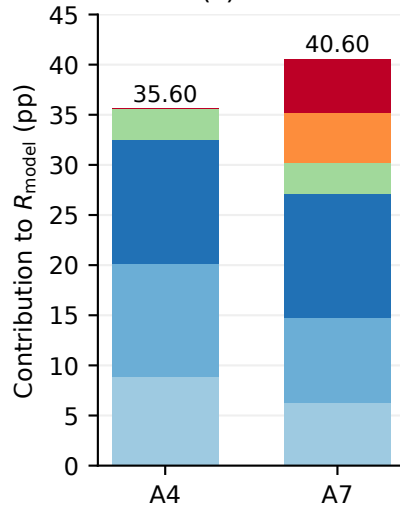
(a) Exact-zero activations (%)



(b) 14M



(c) 70M



Legend: QKV (light blue), MLP up (medium blue), MLP down (dark blue), Attn. output (green), QK (orange), PV (red).